

Investigation of the aberrant record from bus GPS receiver onboard FORMOSAT-7/COSMIC-2 satellite constellation in low Earth orbit

Tzu-Hsun Kao^a, I-Te Lee^{a,b,c,*}, Chen-Joe Fong^d, Jann-Yenq Liu^{a,e}, Ming-Shong Chang^d

^a Department of Space Science and Engineering, National Central University, Taoyuan, Taiwan

^b Advanced Research Center for Earth Sciences, National Central University, Taoyuan, Taiwan

^c Central Weather Bureau, Taipei, Taiwan

^d Taiwan Space Agency, Hsinchu, Taiwan

^e Center for Astronautical Physics and Engineering, National Central University, Taoyuan, Taiwan

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Abstract

The FORMOSAT-7/COSMIC-2 (F7/C2) satellite constellation, a joint Taiwan-US collaboration mission, was launched on 25 June 2019. Following the FORMOSAT-3/COSMIC, the F7/C2 mission aims to provide global users the next-generation global navigation satellite system (GNSS) radio occultation data. The mission consists of six identical small satellites with an inclination angle of 24°, launched to the parking orbit of 720 km, and then transferred to the mission orbit of 550 km. Each satellite bus is equipped with two commercial off-the-shelf global positioning system receivers manufactured by Surrey Satellite Technology Ltd. for positioning, navigation, and timing functions. This receiver generates “GPS Receiver (bus GPSR) Warm Start Requested (WSR)” messages after commissioning, which could be considered as a kind of single event effect (SEE). More than 7000 WSR events have been recorded in 3 years after launch, providing another opportunity to investigate the SEE with the commercial off-the-shelf bus GPSR. In addition, the F7/C2 mission is the first satellite constellation that allows scientists to investigate the SEE using the same constellation at different altitudes of Low-Earth orbit. This study investigates and discusses the cumulative rate, distribution of WSRs, and their relationships between altitude and geomagnetic field. The statistical method is applied to define a warning area of the high probability occurring the SEE of this kind of component based on the geomagnetic field, whose corresponding strength is around 21,200/20,550 nT and close to the South Atlantic Anomaly area. Moreover, the relationships with space weather, including the disturbance of the Earth’s magnetic field (Kp index) and the solar activity (F10.7 index), are also analyzed and reported.

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1. Introduction

The first artificial satellite, Sputnik 1, was launched into the space in 1957. After that, more and more satellites are deployed by various countries into different kinds of orbits. Today, more than 15 thousand artificial satellites are orbiting around the Earth (<https://www.unoosa.org/oosa/>

* Corresponding author.

E-mail addresses: peterhkao@gmail.com (T.-H. Kao), itlee@jupiter.ss.ncu.edu.tw (I-Te Lee), cjfong@tasa.org.tw (C.-J. Fong), jyliu@jupiter.ss.ncu.edu.tw (J.-Y. Liu), mschang@tasa.org.tw (M.-S. Chang).

osindex/), establishing the link between the space environment and human life. With the improvement in space technology, satellites now serve global navigation and positioning, remote sensing, as well as satellite communication in our daily lives. The space environment is a near-vacuum area with a low density of plasma, electromagnetic radiation, magnetic fields, and neutrinos (Horneck, 2015). Furthermore, the charged particles in the ionosphere, thermosphere density, geomagnetic field, and solar activity should be considered in low-Earth orbit (LEO) (Bourdardie and Xapsos, 2008; Nwankwo et al., 2020; Kataoka et al., 2022; Zhang et al., 2022; Gopalswamy et al., 2023). Satellites often encounter energetic particles and radiation hazards while orbiting in the space environment, which may lead to software errors or hardware damages (Campbell et al., 1992; Baker, 2000; Harboe-Sorensen et al., 2001; Lu et al., 2019; Ji and Wang, 2019). These energetic particles can also be detected through several star cameras (Toldbo et al., 2022; Pakhotin et al., 2022). This kind of phenomenon is known as single event effects (SEE), including single event upset, single event latch-up, and single event burnout (Sexton, 2003; Koons and Fennell, 2006). In general, a single event upset is a software error due to one or multiple bits of latches, memory devices, and sequential logic being inverted, which does not affect the regular operation of the components (e.g., Noeldeke et al., 2021). By contrast, a system restart is required for the single event latch-up, a type of hardware failure, which is mostly caused by energetic particles passing through specific circuits to draw an abnormally high operating current (e.g., Tomioka et al., 2016). Meanwhile, single event burnout is the most serious case among hardware failures, mostly caused by the heavy ions that produce the high-current state inside the device (e.g., Albadri et al., 2006; Shiba et al., 2017). Therefore, components of a satellite are comprehensively tested and evaluated for fully operating in the hazardous space environment.

FORMOSAT-7/COSMIC-2 (F7/C2) is an international collaboration program between the Taiwan Space Agency (TASA), formerly the National Space Organization of Taiwan, and the National Oceanic and Atmospheric Administration of the United States (Cook et al., 2013; Fong et al., 2019). As a follow-on mission of FORMOSAT-3/COSMIC (F3/C) (Fong et al., 2009; Chu et al., 2021), it continues to provide global users the next-generation global navigation satellite system (GNSS) radio occultation (RO) data for meteorology, ionosphere, climatology, and space weather (*c.f.* Liu et al., 2015; Chen et al., 2021; Liu et al., 2022). F7/C2 is an operational constellation that consists of six identical small satellites, named F701, F702, F703, F704, F705, and F706 in this study. The payloads onboard each F7/C2 are the Tri-GNSS Radio Occultation System, Ion Velocity Meter, and Radio Frequency Beacon (Curiel et al., 2013). Compared with F3/C, the orbital inclination angle of F7/C2 is 24° in order to acquire more observations at middle and lower latitudes. On 25 June 2019, the six satellites were launched to the circular

parking orbit at 720 km altitude, and transferred to the circular mission orbit at 550 km altitude, sequentially to a planned schedule. Eventually, TASA took about 20 months to complete the constellation deployment. These satellites were deployed into 6 separated orbital planes of the right ascension of the ascending node at 245°, 360°, 70°, 307°, 125°, and 184°, respectively (Weiss et al., 2022). In order to record the satellite's health conditions and unexpected events, wild kinds of messages will be written into logs of the on-board computer (OBC). After F7/C2 was released into the space environment, the logs appear many times of the "GPS Receiver Warm Start Request (abbreviated as WSR here and after)" from a particular element, the bus Global Positioning System Receiver (bus GPSR). The WSR indicates that the bus GPSR is forced to restart due to an unknown reason. The bus GPSR uses the commercial off-the-shelf (COTS) component manufactured by Surrey Satellite Technology Ltd. to provide the positioning, navigation, and timing functions to the spacecraft. It had completed the in-flight evaluation test onboard the SSTL UK-DMC-2 satellite launched in early 2009 (Boland et al., 2005; Ebinuma et al., 2005; Baker, 2006; Unwin et al., 2009). There are two bus GPSRs (one primary and one redundant) with corresponding antennas onboard each F7/C2 satellite, named GPS00 and GPS01. The bus GPSRs are located inside the spacecraft's main bus and their antennas are placed on the zenith side as indicated in Fig. 1.

Three years after launching, 7488 WSRs have been recorded from all six satellites, which means around 208 records per month. Lee et al. (2014) reported 385 reboot/reset (RBS) events of the F3/C from 15 April 2006 to 31 May 2011 that only about 6 times per month. Both WSR and RBS are considered as SEE events; therefore, the large number of WSRs might be abnormal, and worth trying to figure out possible causes. The record of F7/C2 provides a great opportunity to investigate the cumulative rate, distribution of WSR, and its relations with the space weather. Moreover, as the F7/C2 satellites change their orbit from the parking orbit to the mission orbit one by one, it offers an ideal opportunity for the first time to study the characteristics of WSR at different altitudes in one constellation mission.

2. Data analysis

The WSR records from 3 July 2019 to 31 August 2022 are extracted from the OBC logs of the six F7/C2 satellites with a total number of 7,488 WSR events. Considering that the OBC logs document only the time of WSR, the System Tool Kit software (<https://www.ansys.com/products/missions/ansys-stk>) developed by Ansys Government Initiatives is applied in this study to calculate the corresponding orbital information. The software can estimate the latitude, longitude, and altitude of a satellite at a given time via its database and the general perturbation data published by the United States Air Force (<https://www.space-track.org/>). In this

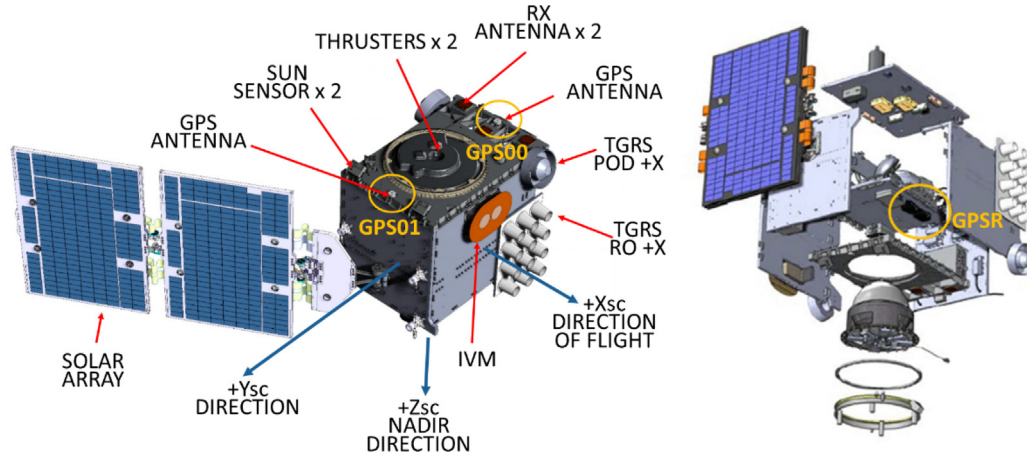


Fig. 1. Configuration of F7/C2 provided by the Taiwan Space Agency. The yellow circles on the left-hand side indicate the antenna of GPS receivers GPS00 and GPS01, while the one on the right-hand side denotes the receivers.

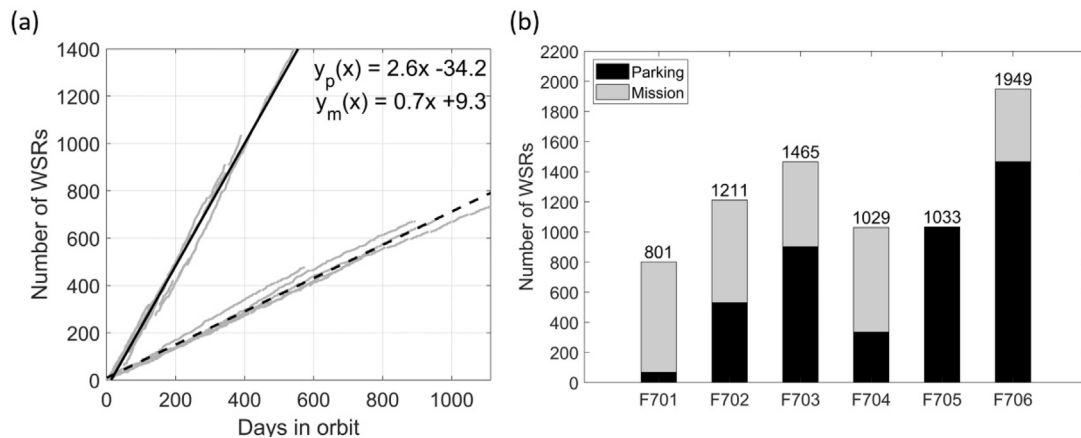


Fig. 2. Analysis of F7/C2 WSR number. (a) Number of WSRs is plotted against the days in orbit. The gray dots indicate the accumulated number of WSR in both orbits. The solid and dashed lines show the linear fitting lines in the parking and operational orbits, respectively. The y_p and y_m indicate the fitting functions of parking and mission orbits. (b) The WSRs accumulated number of each F7/C2 satellite in this study period. The black and gray bars indicate the WSR number in the parking and mission orbits, respectively.

study, the WSR records are separated into parking (4336 times) and mission (3152 times) orbits, since the F7/C2 was launched to the parking orbit first, and then transferred to mission orbit with the order of F701, F704, F702, F703, F705, and F706 (more detailed descriptions see Weiss et al., 2022). Fig. 2a shows the accumulated number of WSRs for the two altitudes. The number of WSRs is plotted against the number of days in the two orbits. The solid and dashed lines are linear fittings of the accumulated number in the parking and mission orbits, respectively. The slopes of these fitting lines can be regarded as the rates of cumulative WSRs. The slopes are 2.6 and 0.7 for the parking and mission orbits, with the former being 3.7 times higher than the latter. Note that the data in the transition altitudes are not included in the analysis. Fig. 2b indicates the WSR accumulated number in each satellite, which generally shows an increasing trend with the F7/C2 decreasing order, except for F705. Due to some unknown reasons, the WSR events were not recorded in F705 OBC logs after descending to the mission orbit.

Therefore, the WSR number of F705 is not consistent with the descending order.

2.1. Spatial distribution

The spatial distributions of WSRs in the parking and mission orbits are shown in Fig. 3a and 3b, respectively. The distributions illustrate clear cut-off edges at around 24°N and 24°S , due to the inclination angle of the F7/C2 orbit. Most of the events are in regions of South America and south Atlantic, while some of these events randomly occurred in other places. Gaillard (2011) reported the mechanism of SEEs is contributed by the interactions of ionized particles and electronic components. The ionized particles will be trapped by the geomagnetic field (Apel et al., 1962; Ray, 1963). Therefore, the spatial distribution of WSRs is further analyzed along the magnetic coordinates (<https://wdc.kugi.kyoto-u.ac.jp/igrf/gggm/index.html>). The distributions depend on the magnetic longitude

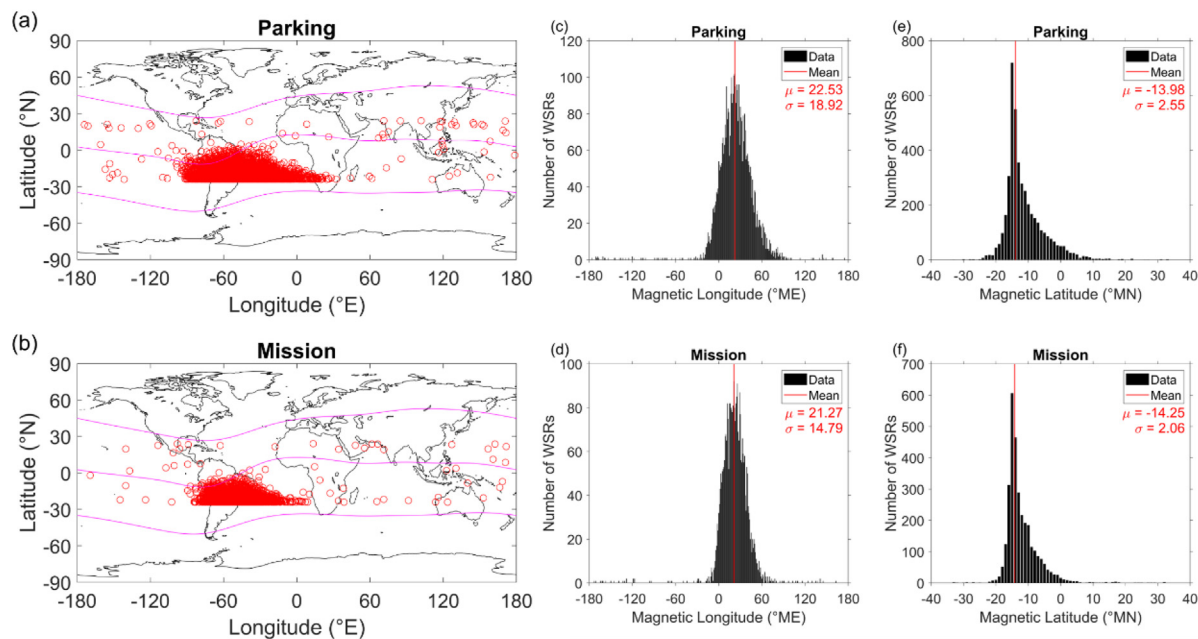


Fig. 3. Analysis of the WSR spatial distribution. (a) and (b) are locations of WSR in the parking and mission orbit, respectively. The magenta lines denote the magnetic latitudes of 40°N, 0°, and 40°S. (c) – (f) are the statistical results along the magnetic longitudinal and latitudinal directions. The mean and standard deviation values are calculated and noted in the charts.

and latitude for the two orbital altitudes are displayed in Fig. 3c–3f. In Fig. 3c and 3d, the magnetic longitudinal weighting centers/standard deviations of the parking and mission orbits are found to be 22.5°ME/18.9° and 21.3°ME/14.8°, respectively. Note that the “M” stands for the magnetic coordinate to discriminate from the geographic coordinate. Fig. 3e and 3f show the magnetic latitudinal weighting centers/standard deviations for the parking and mission orbits are 14.0°MS/2.6° and 14.3°MS/2.1°. The corresponding geographic weighting center of the parking orbit and the mission orbit are at (22.8°S, 49.0°W) and (23.2°S, 50.2°W), respectively, as shown in Table 1.

These WSRs may be related to the local geomagnetic field strength at the satellite positions. In this study, the International Geomagnetic Reference Field 13th (IGRF-13) model (Alken et al., 2021) is applied to investigate the relationship between WSRs and the Earth’s magnetic field. As Fig. 4 shown, 95% of WSRs occurred in the area where the magnetic strength is less than 21,200 and 20,550 nT in the parking and mission orbits, respectively. Fig. 5 illustrates the WSRs spatial distribution of both orbits combined with the geomagnetic field strength accord-

ing to the IGRF-13 model in the selected region (–50–20°N, –120–40°E). The proposed geomagnetic field strength of 21,200 and 20,550 nT in both orbits are denoted as borders, and the 32,000 nT suggested by De Santis et al. (2013) to determine the region of the South Atlantic Anomaly (SAA) on the ground is also appended. The SAA is the area that has the lowest geomagnetic field on the Earth, where the lower boundary of the inner radiation belt could reach an altitude of only a few hundred kilometers (Kurnosova et al., 1962; Vernov et al., 1967; Ye et al., 2017). It’s worth noticing that there are accumulations of WSRs eastward of the proposed 21,200/20,550 nT boundaries. For the parking orbit, there are 5.56% (241 WSRs) in the east and 2.33% (101 WSRs) in the west with a total of 4,336 WSRs. For the mission orbit, there are 5.17% (163 WSRs) in the east and 1.78% (56 WSRs) in the west with a total of 3,152 WSRs. Both altitudes show a comparable result that more than 3% of events are located outside the eastern border. Meanwhile, similar patterns are also observed through SAMPEX, Swarm, DMSP, and DORIS/Jason-1 observations. (Willis et al., 2004; Lemoine et al., 2006; Jones et al., 2017; Anderson et al., 2018; Toldbo et al., 2022).

Table 1
The weighting centers (WC) and standard deviations (SD) of WSRs spatial distribution.

Orbit		Geomagnetic Coordinate	Geographic Coordinate
Parking (720 km)	WC	(14.0°MS, 22.5°ME)	(22.8°S, 49.0°W)
	SD	(2.6°, 18.9°)	–
Mission (550 km)	WC	(14.3°MS, 21.3°ME)	(23.2°S, 50.2°W)
	SD	(2.1°, 14.8°)	–

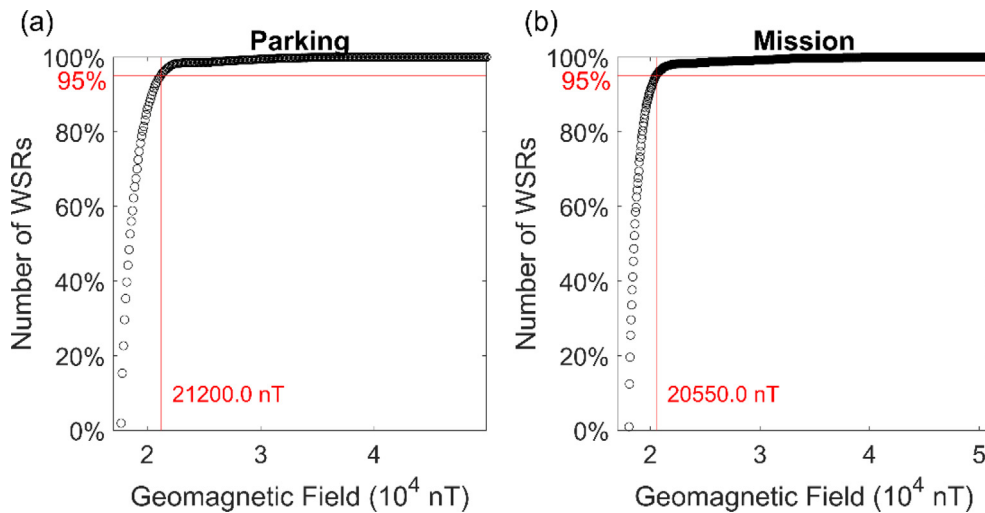


Fig. 4. The accumulated WSRs are based on the magnetic field strength in the parking (a) and mission orbits (b).

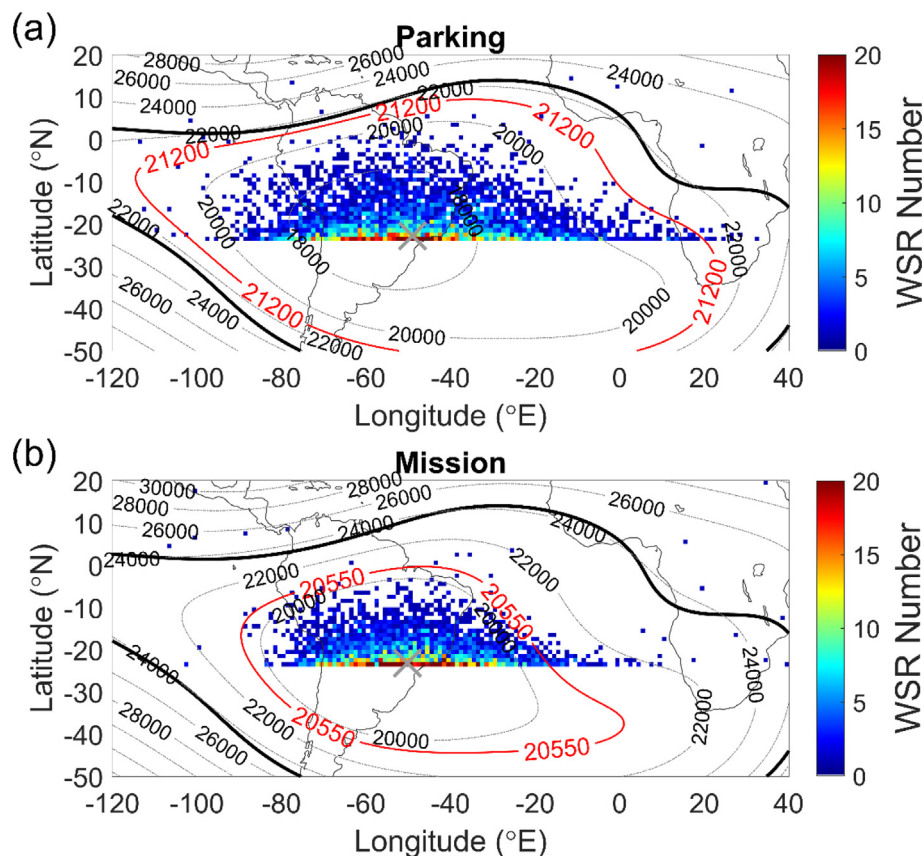


Fig. 5. The spatial distribution of WSR number is shown in color maps and combined with the geomagnetic contour (gray) produced by the IGRF-13 in the parking (a) and the mission (b) orbits in 2020. The red lines indicate the magnetic strength equals to 20,550 nT (a) and 21,200 nT (b) at 550 and 720 km in 2020, respectively. The bold black line denotes the 32,000 nT at ground in 2000 as the boundary of SAA reported by [De Santis et al. \(2013\)](#).

2.2. Space weather condition

The number of WSRs is classified with the Kp index and the solar flux index (F10.7 index), which are commonly used to evaluate geomagnetic conditions and solar activities ([Bartels, 1949](#); [Allen, 1982](#); [Bartels, 2013](#); [Tapping,](#)

[2013](#)). The Kp and F10.7 indices are all employed from the GeoForschungsZentrum (GFZ) database ([Matzka et al. 2021a](#)). It was a quiet/low geomagnetic condition/solar activity at the beginning of this study period when most satellites were orbiting in the parking orbit. In order to cover more solar activity conditions and avoid differences

in orbital altitudes, this analysis only involves WSRs in the late period, which are occurring in the mission orbits. The Kp index is reported every 3-hour interval based on 13 geomagnetic observatories around the globe in 27 levels (Matzka et al., 2021b), and the F10.7 is registered once a day in solar flux unit ($1 \text{ SFU} = 10^{-22} \text{ Wm}^{-2}\text{sHz}^{-1}$) (Tapping, 2013). To compare the possibility of occurring WSRs under different space weather conditions, an incidence rate is defined as the number of WSRs occurring per satellite within a period of time ($\#/\text{sat.}/\text{period}$).

Fig. 6a shows the statistical result associated with the Kp value, the 3-hour counts at each Kp value in this study period, and its corresponding incidence rate in 3 h. The incidence rate of Kp is calculated based on the averaged WSR number of all satellites per 3 h ($\#/\text{sat.}/3\text{-hr}$), which is consistent with the time resolution of the Kp index. Most WSRs occur during quiet geomagnetic conditions ($K_p \leq 4$). The incidence rate generally appears stable under quiet geomagnetic conditions, but less stable under active conditions. The top panel of Fig. 6b also provides

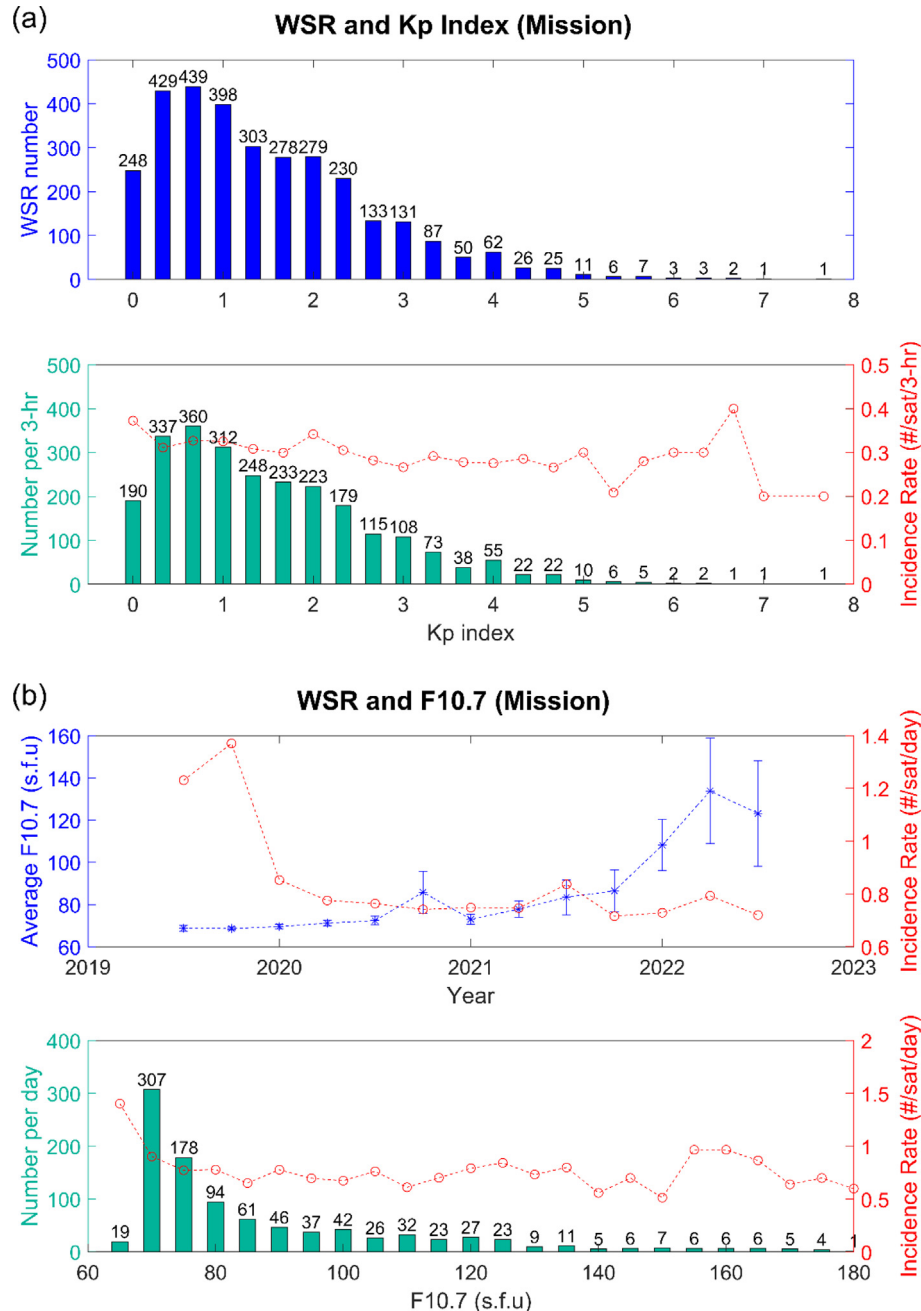


Fig. 6. The relationship between WSRs and Kp index as well as solar flux index (F10.7 index). (a) The top panel reveals the accumulated number of WSRs on each Kp value, and the lower panel illustrates the number per 3-hour of each Kp value and incidence rate. (b) Similar to (a), with the F10.7 index (blue asterisk) and the corresponding incidence rate (red circle) every 3 months on the top panel. The F10.7 is divided into several groups with 5 SFU intervals in the lower panel.

the variation of the 3-month averaged F10.7 index in each season and the corresponding WSR incidence rate. The incidence rate of F10.7 is calculated based on the averaged WSR number of all satellites per day ($\#/\text{sat.}/\text{day}$). From 2019 to 2022, the F10.7 index gradually increased, revealing that solar activity is progressively more active in this solar cycle 25. Although the F10.7 increases, the incidence rate does not reveal clear relationships. The higher incidence rates in the first two seasons might result from the fewer WSRs occurring in mission orbit. The lower panel of Fig. 6b denotes the number of days in each F10.7 group within 5 SFU intervals and the corresponding incidence rate. The incidence rate also registers stable under the solar quiet days (F10.7 less than 140 SFU), and less stable at the active days. Note that the incidence rates being less stable in the active geomagnetic period and high solar activity day is due to the less WSR and time during this investigation period.

3. Discussion

In Fig. 2, the WSR accumulates at a higher rate in the parking orbit than that in the mission orbit. Due to the protection of the geomagnetic field, satellites in the lower orbits are less likely to be influenced than those in higher orbits (Harboe-Sorensen et al., 2001; Benton and Benton, 2001; Kamide, 2001; Iucci et al., 2005). This may be due to the dissimilar flux of the trapped particles at different altitudes. Martines (2011) reported that as the altitude increases, the flux of trapped protons and electrons are also increasing. The higher rate represents that more WSRs occurred on each F7/C2 satellite while they orbited at higher altitudes, which agrees with the previous studies. Furthermore, the SEE events of the OBC on FORMOSAT-2 (F2) at about 890 km altitude and FORMOSAT-3/COSMIC at about 800 km altitude, reported by Lee et al. (2014), are referenced to make further comparisons. The average monthly records of SEE are 0.15 ($\#/\text{sat.}/\text{month}$) for F2 and 1.07 ($\#/\text{sat.}/\text{month}$) for F3/C, but that of F7/C2 is nearly 33.73 ($\#/\text{sat.}/\text{month}$). The number is about 224.9 and 31.5 times more than F2 and F3/C, respectively. The average SEE number of F3/C is larger than F2. This might be because F2 is well-shielded with aluminum and employs highly reliable electrical, electronic, and electromechanical parts, but F3/C does not (Lee et al., 2014). Although the structure of F7/C2 is well-shielded and its altitude is lower than F2 and F3/C, the number of SEE is expected to be lower than these two missions. However, the number of WSRs is much larger, which might be due to the bus GPSR being a COTS component and can be expected to be more sensitive to radiation (Underwood and Oldfield, 1998; Monreal et al., 2019). Fortunately, it would not cause severe problems for the satellite operation. Due to the bus GPSR is an independent system, not associated with other mission payloads. Also, the high number of WSRs provides a good opportunity to conduct detailed analyses.

According to the spatial analysis (Fig. 3), weighting centers and standard deviations of WSRs are calculated and reported in Table 1. The weighting centers are at (22.8°S, 49.0°W) and (23.2°S, 50.2°W) for each orbit. The standard deviations in latitudinal/longitudinal are 2.6°/18.9° and 2.1°/14.8° in the parking and mission orbit, respectively. Based on the standard deviations, locations of WSRs are more concentrated over South America in the mission orbit than that in the parking orbit. The weighting centers in both altitudes do not change a lot, only about 1° difference. The standard deviation is smaller in the mission orbit, which might result from the wider magnetic tube at higher altitudes, and further suggests that the distribution of WSRs follows the structure of the geomagnetic field. The lower geomagnetic field strength may lead the charged particles to penetrate down to a lower altitude (Pilipenko et al., 2006). Hastings and Garrett (2004) reported that satellites in low Earth orbit may be affected by the inner radiation belt over the SAA area where the weakest geomagnetic field on Earth is located. Meanwhile, the pattern shown in Fig. 3 generally agrees with the SAA area shown in Haines et al. (2003) by investigating the BlackJack receiver onboard the Jason-1 mission at 1300 km altitude. The SAA area defined by Jason-1 is wider than that defined by the WSRs might be due to the satellite orbiting at a higher altitude but with weaker magnetic field strength. This would further support that the longitudinal standard deviation of 18.9° is greater in the parking orbit than the 14.8° in the mission orbit. Therefore, the weighting center of the WSRs distribution could be considered the center of the SAA. The shift of the SAA could also be interpreted through this study. Jones et al. (2017) used the SAMPEX satellite to estimate the secular drift of the SAA and reported the center of SAA was located at (23°S, 47°W) in 2011. In 2019–2022, the weighting centers of WSRs are (22.8°S, 49.0°W) and (23.2°S, 50.2°W) in the parking and mission orbit, respectively. Comparing these locations, the longitudinal location differs from Jones et al. (2017) is about 2°, which agrees with their result that the motion of SAA is 0.2 degrees per year westward. This once again shows the westward drift of the center in both orbits, which also meets the results of Grigoryan et al. (2008) and Casadio and Arino (2011). However, the latitudinal shift cannot be yielded due to the inclination angle limitation at 24°.

Previous studies have defined the SAA as the region that has a magnetic field strength of less than 32,000 nT on the ground in 2000 (De Santis et al., 2013; Pavón-Carrasco and De Santis, 2016). Lee et al. (2014) discovered that the position of F3/C RBS is related to the geomagnetic field and mostly occurred over the SAA where it is less than 22,700 nT, but not denoted the corresponding altitude and year. In this study, both the previous boundaries of 32,000 nT on the ground and 22,700 nT are larger than the distribution of WSRs (Fig. 5) in both orbits. As the result shown in Fig. 4, 95% WSRs of these two orbits occurred at locations where the geomagnetic strength is less

than 21,200 nT and 20,550 nT. The other 5% located in other areas might be due to random ionized particles hitting, discharges, magnetic fluctuations, or radiations. Based on the analysis of both spatial distribution and geomagnetic model, the magnetic field strength of 21,200 nT and 20,500 nT would indicate new reference boundaries of SAA, as well as a possible warning area of occurring SEEs on these kinds of COTS components at 720 km and 550 km altitudes. The magnetic field strength of the SAA boundary reported here is weaker and the area is smaller than that suggested by the previous research.

In Fig. 5, it's clear to see that there are accumulations of WSRs eastward of the proposed 21,200/20,550 nT boundaries. More than 3% of events occurred outside the eastern border, which generally agrees with previous studies (Willis et al., 2004; Lemoine et al., 2006; Jones et al., 2017; Anderson et al., 2018; Toldbo et al., 2022). This phenomenon might be due to the small-scale structure of Earth's geomagnetic field not described by the IGRF model. The IGRF model is a mathematical model with a set of spherical harmonic coefficients (Aiken et al., 2021). The coefficients are generally calculated by ground-based magnetometers and satellite measurements, including Swarm, Ørsted, CHAMP, SAC-C, Cryosat-2, and CSES. It can describe the large-scale and time-varying portion of Earth's internal magnetic field from 1900. In other words, some small structures of the geomagnetic field might not be surely reproduced by this model.

On the other hand, the change in space weather conditions may also affect satellite anomalies. Previous studies conducted mostly based on the geostationary satellites and pointed out that most of the satellite anomalies were highly correlated to active geomagnetic conditions and high solar activities (Allen et al., 1989; Fennell et al., 2001; Belov et al., 2004; Iucci et al., 2005; Likar et al., 2012). However, the orbital altitude of the F7/C2 is lower than the geostationary satellites, which might result in the WSR incidence rate not yielding a positive correlation with space weather conditions. Another reason for the unclear correlation would be the absence of severe space weather and a short period of analysis time. In addition, Shen et al. (2021) analyzed the F3/C orbiting at 800 km altitude and concluded that the LEO satellite anomalies were anti-correlated to solar activities and geomagnetic conditions under certain space weather conditions (K_p less than 6, Daily Sunspot Wolf number < 120). They explained that the altitudes of the Earth's upper atmosphere would expand to higher altitudes during active geomagnetic conditions and high solar activities (Lühr and Liu, 2006; Wang et al., 2022; Fang et al., 2022), which results in higher neutral atmospheric density in the SAA region. This would lower the flux of high-energy protons during high solar activity and geomagnetic conditions (Farley and Walt 1971; Miyoshi et al. 2000; Qin et al. 2014; Zou et al., 2015). Since F7/C2 is orbiting at a lower altitude than F3/C, this effect will be more dominant to protect the lower LEO satellites from hitting by high-energy protons.

Meanwhile, there's no severe space weather during the period. Both the aforementioned might be the possible reasons to explain that the WSR incidence rate and the space weather do not reveal obvious correlations. After accumulating more WSRs under higher geomagnetic/solar activity in the future, it would have a better opportunity to carry out the relationship between WSRs and the space weather.

4. Conclusion

Since F7/C2 records a large number of WSRs on bus GPSR, it gives scientists a good opportunity to investigate the SEE with the COTS component. In addition, the F7/C2 mission has six identical satellites and transfers from the parking orbit (720 km) to the mission orbit (550 km), which is the first satellite constellation that could be used to investigate the SEE with the same constellation in different altitudes. In this study, the cumulative rate, distribution of WSRs, and the relationships between WSRs as well as altitude, geomagnetic field, and space weather are investigated and discussed.

The result shows that the cumulative rate is larger while the F7/C2 satellites are orbiting at a higher altitude (Fig. 2). The distribution of WSR recorded by all six satellites is majorly over the South America and the South Atlantic region. The spatial distribution analysis (Fig. 3) provides the weighting center and standard deviation in both altitudes listed in Table 1. According to 95% of accumulated WSRs and their corresponding geomagnetic field strength, it suggests new reference boundaries at around 21,200 nT and 20,550 nT in the parking and mission orbits respectively for the SAA and circles of the SEE warning areas of these kinds of COTS components in the low-Earth orbit. In addition, there is an accumulation eastward of WSRs spatial distribution, which is possibly due to the IGRF model that might not describe the small structure of the geomagnetic field. In the end, both the K_p and F10.7 indices do not reveal clear relationships with the WSR incidence rate, which suggests the SEE would possibly not be significantly affected by the space weather in the lower Earth orbit or the longer research period is needed. These investigations of WSRs are essential for satellites or CubeSats using this kind of COTS component.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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NSPO-S-110039. The data of F10.7 index and Kp index are available at GeoForschungsZentrum (GFZ) database (<https://www.gfz-potsdam.de/en/section/geomagnetism/data-products-services/geomagnetic-kp-index>). The IGRF model was provided by the World Data Center for Geomagnetism, Kyoto (<https://wdc.kugi.kyoto-u.ac.jp/>). The authors also thank the team of TASA (formerly NSPO) for operating and controlling the FORMOSAT-7/COSMIC-2. TZK thanks Ms. Liu, Tien-Chi at National Central University for giving further suggestions.

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